

Tech and Tools, Mathematica Lab, Part I

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For 2:00-4:00pm, Tuesday, 8 July 2003

Part I is due after the lab at 6:00pm

1. Try to familiarize with the notebook interface of Mathematica. Read the first few sections of the Mathematica Book. Use the help system to find out the command `N[ ]`. Type some text (as document) and enter a command. Change the font/color of the text. Change the stylesheet. Try cut and paste, etc.
2. We want to be familiar ourselves to arithmetics in Mathematica. We can use it as a pocket calculator. Consider entering the following mathematical expressions to the system and watch the reaction of the system:

$$\begin{array}{ccc} 1 + 2, & 2 - 3 \times 5 - 1/2, & 2/3 - 0.5, \\ \sqrt{4}, & i^2, & 2^{10}. \end{array}$$

3. All symbolic systems can handle arbitrarily large integers without overflow. Try the following:

$$2^{100}, \quad 30!, \quad 123456789 * 987654321.$$

4. Mathematica can do arbitrarily precise floating-point computation. `N[ ]` converts an exact number into an arbitrary precision floating-point value. Find the numerical values of the following numbers accurate to 30-digit precision (quad precision):

$$\begin{array}{ccc} \pi, & e, & \sqrt{2}, \\ 355/113, & \sin \frac{\pi}{7}, & e^{\pi\sqrt{163}}. \end{array}$$

In Mathematica, make the assignment `pi = N[Pi,30]`. What is the digits of precision of `2*pi`, and what is `2.0*pi`? Why the second expression displays you only six digits? How to remove the symbol `pi` from the system when you have finished using it?

5. In this exercise we like to see how Mathematica deal with symbols. What happens when the following mathematical expressions are typed into symbolic systems:

$$\begin{array}{ccc} \pi, & x + y, & x - x, \\ 2x \times x + 1 & (1 + x)^2 & \sin \frac{\pi}{2}. \end{array}$$

Note that π should be typed in as the constant `Pi`, not a numerical approximation.

6. Consider the cubic equation

$$f(x) = x^3 + 6x^2 + 9x + 1 = 0.$$

- (a) Make a plot to see how many real roots this equation has. Read from the plot the rough values of the solutions.
- (b) Solve the equation numerically.
- (c) Solve the equation symbolically. The resulting expressions are complicated with imaginary number i . But we know the solution has to be real (from the plot). Simplify the expressions if you can.
7. (This is an extra bonus problem if you have time). Plot the curve (using `ParametricPlot[ ]`) given by $r = \theta$ (polar coordinates). Plot two curves given by $r = \theta$ and $r = 1$ on the same plot, using two different colors.
8. Mathematica is good at calculus. Compute the following (partial) derivatives with Mathematica.

$$\begin{array}{lll} \frac{d}{dx} \sin x, & \frac{d}{dx} x^x & \frac{\partial}{\partial y} (x^3 - 3xy^2), \\ \frac{d^2}{dx^2} \cos^{-1} x, & \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (x + iy)^3, \quad i^2 = -1, & \frac{d}{dx} f(x^2). \end{array}$$

9. Differentiation can be done for any combination of algebraic expressions. But indefinite integration is much harder. Not every expression can be integrated.

(a) Try the following integration by Mathematica.

$$\int x \, dx, \quad \int \sin x \, dx, \quad \int (1 + e^x)^{-1} \, dx.$$

(b) And some more complex (type in very very carefully):

$$\int \frac{x \{ 2x e^{3x^2} [x - (1 + x + 2x^2 + 2x^3) \log(1 + x)] + [x^2 e^{2x^2} - \log^2(1 + x)]^2 \}}{(1 + x) \{ -x^2 e^{2x^2} + \log^2(1 + x) \}^2} \, dx.$$

(c) Try the ratio of polynomial function.

$$\int \frac{6x^5 + 6x^4 - 8x^3 - 18x^2 + 8x + 8}{x^6 - 5x^4 - 8x^3 - 2x^2 + 2x + 1} \, dx.$$

(d) Try

$$\int \frac{x}{1 + e^x} \, dx$$

10. Find the five (numerical) roots of the equation

$$x^5 + 5x^4 + 4x^3 + 3x^2 + 2x + 1 = 0.$$

Solve the equation again with a higher precision, for example, 30 digits of accuracy.

11. Solve the set of second order equations

$$\begin{array}{l} y^2 + x^2 - y - 3x = 0, \\ y^2 - 6xy - x^2 + 11y + 7x - 12 = 0. \end{array}$$

Verify that the solutions are correct by substituting the solution back.

12. Define a Mathematica function $f(x) = 1/(x^2 + 1)$, and make a plot of it. Define the integer factorial function $n!$ in Mathematica. Define the greatest common divisor function $\text{gcd}(a, b)$ of two integers; compare with build-in function $\text{GCD}[\dots]$. Generalize your gcd definition so that it works for 1, 2, 3 or any number of arguments.

Note: greatest common divisor integer c , of a set of integers $a_1, a_2, a_3, \dots$, is the largest number such that c divides a_i for all i exactly. In other words, there exists an integer d_i such that $a_i = cd_i$. Hint: recursive definition would be best.

13. [Really optional] The three-dimensional laplacian operator in cartesian coordinates (x, y, z) is given as

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Using Mathematica, find the same operator in spherical polar coordinates (r, θ, ϕ) , related to cartesian coordinates by

$$\begin{aligned}x &= r \sin \theta \cos \phi, \\y &= r \sin \theta \sin \phi, \\z &= r \cos \theta.\end{aligned}$$

This involves viewing f (in $\nabla^2 f$) as a function of r, θ , and ϕ , using the chain rule to compute partial derivatives. [N.B. we want to find $\hat{\nabla}^2$ such that $\nabla^2 f(x, y, z) = \hat{\nabla}^2 \hat{f}(r, \theta, \phi)$ where $\hat{f}(r, \theta, \phi) = f(x(r, \theta, \phi), y(r, \theta, \phi), z(r, \theta, \phi))$.]